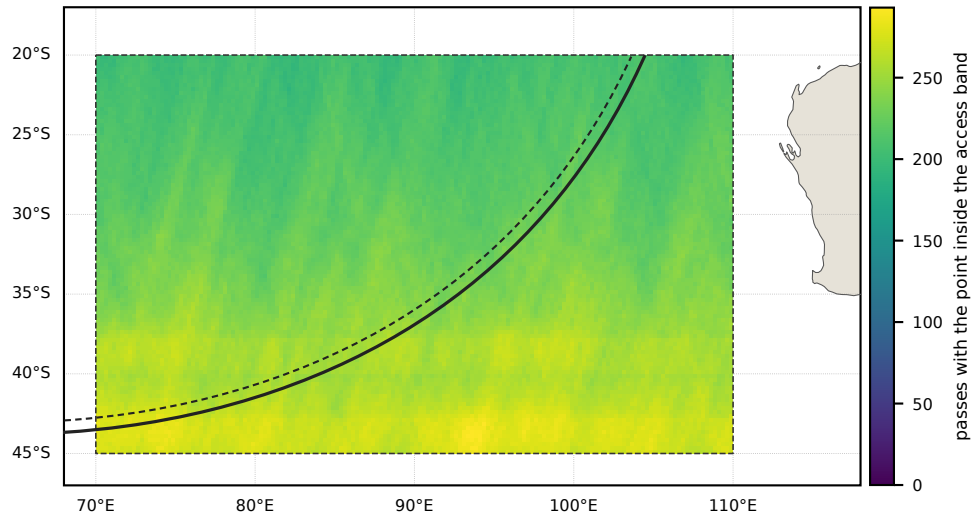
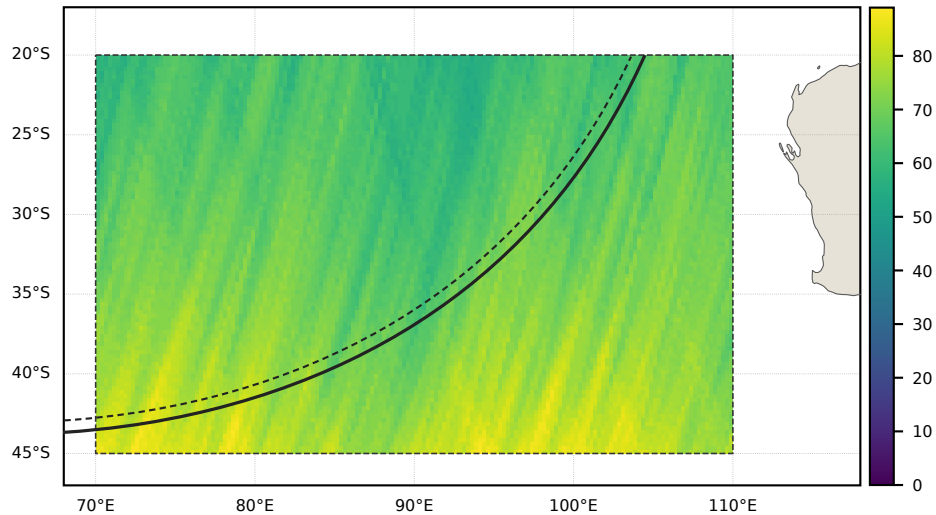
# Theoretical satellite imaging access over the 6th and 7th arcs, 7 Mar 16:42 UTC to 16 Mar 00:00 UTC 2014

Count of satellite passes in which each point lay inside the sensor's possible field of view. Solid line: 7th arc (00:19). Dashed line: 6th arc (00:11). Dashed box: study area 20–45 S, 70–110 E.

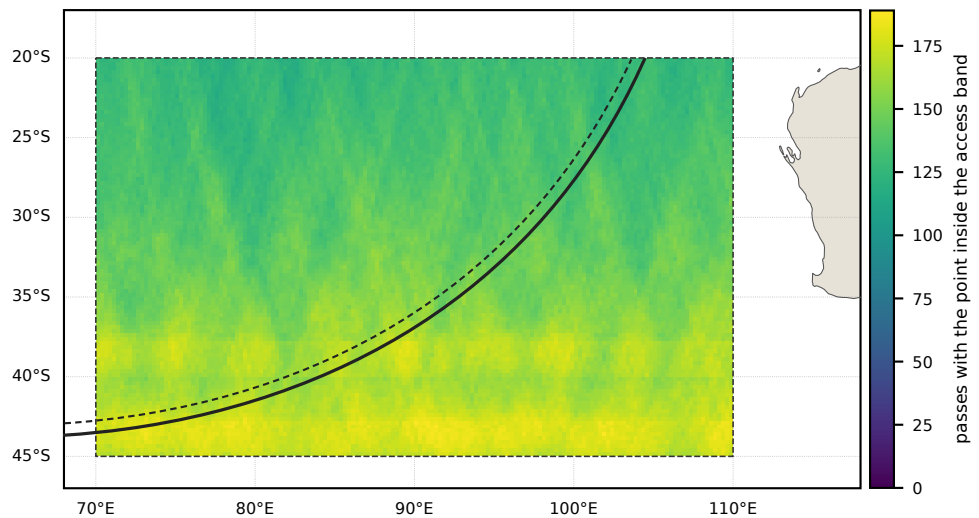
a Sensors able to resolve a 1 m object (27 optical in daylight, any off-nadir angle; 14 radar)



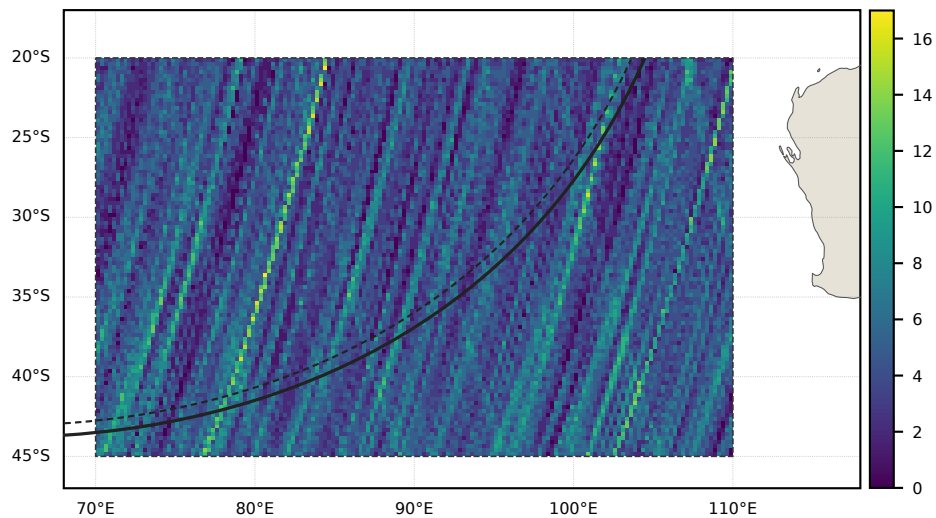

b Same sensors, optical pointing limited to 30° off-nadir




c Radar satellites only (20; day or night, through cloud), full incidence range




d Nominal swaths of the 51 optical sensors of 5 m or better (no off-nadir pointing




Plain-language note: This map does not show where images were taken. It shows how often a satellite with a camera or radar good enough to see a 1 m object passed within reach of each point, with its camera turned as far as the operator allows (a), turned no more than 30° from straight down (b), or radar only (c). Panel d shows the strips directly below the optical satellites of 5 m or better, i.e. what they image with no special pointing. Optical passes count only in daylight (sun more than 10° above the horizon). No agency has said that any of these satellites took images here in this period; the Chinese constellation was re-tasked from 10–11 Mar for the Gulf of Thailand and the Malacca Strait, not this area.

Technical: SGP4 from Space-Track element sets nearest in epoch (public bulk export; 6,568 sets, 77 objects, 1–28 Mar 2014) plus amateur sets for USA 186, USA 245, Lacrosse 5, SAR-Lupe 3 and 5 (McCants classfd, 19 Mar 2014 capture); 10 s steps; TEME→ECEF by GMST only. Access band: optical = ground offset at the operator's maximum off-nadir angle (per satellite, 20–60°) + half swath; SAR = ground offsets at minimum and maximum incidence on the look side(s). Validation: THEOS 24 Mar closest approach 03:08:00 vs 03:07:56 published, incidence 26.91 vs 26.98°; WorldView-2 16 Mar 04:15:40 vs 04:16. Not included: IGS, Helios, Ofeq, SAR-Lupe 1/2/4, USA 224, Topaz (no March 2014 elements); 28 Flock-1 Doves (commissioning); weather sensors of 250 m or coarser. Grid 0.25°.